

AI Practice Test

Practice Test 1 Link:

<https://wayground.com/join?gc=12314441&source=liveDashboard>

Practice Test 2 link: <https://wayground.com/join?gc=48457545&source=liveDashboard>

Practice Test 3: <https://wayground.com/join?gc=16605897&source=liveDashboard>

Practice Test 4: <https://wayground.com/join?gc=39695049&source=liveDashboard>

Practice Test 5: <https://wayground.com/join?gc=12906185&source=liveDashboard>

Practice Test 6: <https://wayground.com/join?gc=03469001&source=liveDashboard>

Final Mock on Wayground:

<https://wayground.com/join?gc=04741065&source=liveDashboard>

Machine Learning Practice Test :

<https://wayground.com/join?gc=16580041&source=liveDashboard>

Summary :

1. Temperature

Definition: Controls the randomness or creativity of an AI model's output.

- **Low temperature (0.2):** Predictable, safe responses.
- **High temperature (0.8–1.0):** Creative, diverse responses.

Example: - Temp 0.2 → "The cat is sleeping." - Temp 0.9 → "The cat leaps across the moonlit garden, chasing shadows."

Use case: - Chatbots: Higher temp → more creative answers. - Summarization: Lower temp → more precise answers.

2. Grounding

Definition: Ensuring AI outputs are factually accurate and tied to trusted sources.

Why needed: Prevents hallucinations and ensures reliable answers.

Example: - Without grounding: "The Eiffel Tower is in Berlin." ✗ - With grounding: "The Eiffel Tower is in Paris." ✓

Domain-specific view: - Grounding ensures AI answers are **correct in a broad domain**. - Medical example: "Explain symptoms of diabetes" → AI uses **general medical knowledge** reliably.

Analogy: Like a doctor who knows **general medicine** and only speaks factual information.

3. Fine-Tuning

Definition: Training a pre-trained model further on a specific dataset to make it specialized.

Example: - Base model: GPT-4 - Fine-tuned: Legal document assistant (understands law-specific terms).

Use case: - Custom chatbots - Domain-specific Q&A - Specialized predictions

Analogy: Training a general athlete to specialize in swimming.

4. RAG (Retrieval-Augmented Generation)

Definition: A method to make LLMs factual by retrieving relevant documents and generating answers using them.

Difference from Grounding: - Grounding = Goal (answers are factual in a domain) - RAG = Technique (retrieves specific information to generate answer)

Domain-specific view: - RAG allows AI to retrieve **specific documents or sources** to answer detailed questions. - Medical example: "What were the lab results for patient X at Hospital Y?" → AI retrieves **the exact hospital record** and generates the answer.

Analogy: - Grounding = Doctor knows general medicine - RAG = Doctor checks the **patient file** before answering a question - Library analogy: Grounding = trusted medical library; RAG = find the exact book and page for your question

5. Quick Summary Table

Concept	Function	Analogy
Temperature	Controls randomness/creativity	Spicy level of answer
Grounding	Ensures factual accuracy across a domain	Doctor knows general medicine / check the textbook
Fine-Tuning	Specializes a model	Train an athlete for a specific sport
RAG	Technique for grounding using retrieval	Find the exact book/page in the library / check patient file

Generative AI Concepts

Concept	Practical Example
Temperature	Chatbots like ChatGPT or customer service AI — adjust creativity vs safety of responses
Grounding	Knowledge mining systems, medical AI assistants — ensures factual answers from trusted domain knowledge
Fine-Tuning	Custom chatbots for law, finance, or healthcare — improves domain-specific understanding
RAG	Enterprise Q&A systems — retrieves documents from company knowledge base to answer questions

Summary of few topics from Generative AI : just for an overview:

Concept	What it means	Real-world scenario	Simple example	Mitigation / How to improve	AI-900 Topic
Temperature	Controls randomness/creativity of output	Chatbot answering vs creative writing	Low temp → “Your bill is \$50” / High temp → creative story	Use low temp for factual tasks, high for brainstorming	Prompt Engineering
Groundedness	Response based on trusted external data	HR bot answering company policy questions	AI answers leave policy using company docs	Provide verified documents, databases	Evaluation
RAG (Retrieval-Augmented Generation)	Fetch data first, then generate answer	Chatbot using internal company files	AI retrieves PDF policy	Connect LLM to	Improving Accuracy

Augmented Generation)			→ then answers	search/vector DB	
Coherence	Logical flow of response	AI giving step-by-step explanation	Answer jumps between unrelated ideas	Better prompting, structured output format	Evaluation
Fluency	Grammar + readability quality	Customer email response	"We assist your request promptly."	Use better prompt + model tuning	Evaluation
Relevance	Whether answer matches question	User asks refund policy, AI talks about shipping	Off-topic response	Improve prompt + retrieval filtering	Evaluation
Hallucination	AI makes up incorrect facts	AI invents policy or fake data	"Your refund is 90 days" (false)	Use RAG, grounding, validation	Risk
Similarity	Matches AI answer to expected answer	Grading AI-generated answers	Compare student answer vs correct answer	Use reference dataset	Evaluation
System Message	Hidden instructions guiding AI behavior	"You are a support agent"	AI responds politely as assistant	Carefully design system prompts	Prompt Engineering
Tokens	Pieces of text used by model	Sentence split into token chunks	"Hello world" → [Hello, world]	Optimize prompts for token limits	LLM Basics
Context Window	How much text AI can remember at once	Long conversation/chat history	AI forgets early part of chat	Summarize or chunk data	LLM Basics
Top-P (Nucleus Sampling)	Chooses from most likely words	AI generating sentence variety	Picks from top 90%	Adjust for creativity vs stability	Prompt Engineering

			probability words		
Content Filtering	Blocks unsafe/harmful content	User asks for harmful instructions	AI refuses unsafe request	Azure safety filters + policies	Responsible AI

Text Analysis:

Concept	Simple Meaning	How it Works	Why it is Used	Example Scenario	Limitation
Text Frequency (Frequency Analysis)	Counts how often each word appears in a document	Tokenize text → count word occurrences	Identify main topic of a document	In reviews, “AI” appears 10 times → document is about AI	Ignores meaning across documents
TF-IDF	Finds important words in a document compared to all documents	TF (word count in doc) × IDF (rarity across docs)	Highlights unique keywords per document	“Copilot” important in Doc A but not common in all docs	More complex than simple counting
Term Frequency (TF)	How often a word appears in one document	Count occurrences	Measures importance inside a single document	“agent” appears 5 times in a document	Common words may dominate
Inverse Document Frequency (IDF)	Measures how rare a word is across all documents	$\log(\text{total docs} / \text{docs containing word})$	Reduces importance of common words	“Microsoft” appears in all docs → low value	Needs multiple documents
Bag of Words (BoW)	Represents text as word counts ignoring order	Convert text → word frequency vector	Input for ML models like spam detection	“free win prize” → feature vector	Ignores grammar & order
N-grams	Groups words together instead of single words	Combines words into pairs/triples	Captures phrase meaning	“artificial intelligence” as one unit	Increases feature size

Stop Word Removal	Removes common words with little meaning	Remove "the", "is", "and"	Improves focus on important words	"the AI system is fast" → removes "the, is"	May remove useful context
Stemming	Cuts words to root form (rough method)	Remove endings like -ing, -ed	Reduce word variations	"running → run", "powered → power"	Can produce incorrect words
Lemmatization	Converts words to correct dictionary form	Uses language rules	More accurate normalization	"running → run", "better → good"	Slower than stemming
Text Normalization	Standardizes text before analysis	Lowercase + punctuation removal	Clean input for processing	"AI." → "ai"	May lose meaning (Mr. Banks issue)
TextRank	Finds important sentences using graph scoring	Sentences linked by similarity → ranked	Used for summarization	Selects key sentences from cloud text	Only extractive summary

Examtopics PDF: https://npowercanada-my.sharepoint.com/:b:/g/personal/geetu_sodhi_npowercanada_ca/IQCwLVBbNwZ1RYIs4RwbkwCBAf9YF8nN2pUnH9MtKUbdUS0?e=4nkb62